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CS 121: Information Retrieval

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Assignment 3 M3: Optimizations Report

Initial testing queries (pre-optimizations)

Query	Report (fastest to slowest)
aaaaa	Good
lol	Good
informatics	Good
information	Good
computer computer computer	Good
artificial intelligence	Good
cristina lopes	Good
I am	Good
machine learning	Good
undergraduate programs	Good
data science	Good
It is	Bad
Computer science	Bad
javascript react nodejs frontend	Bad
uci cs graduate admissions	Bad
uci computer science	Bad
student and teacher	Bad
To be or not to be	Bad
the best professor at uci	Bad
This and that	Bad

master of software engineering program requirements application deadline	Bad
how do i apply for the master of software engineering program at UC Irvine	Bad

Reasoning for “bad” queries:

We believe the main reasons the “bad” queries had slow runtimes were that we weren’t doing duplicate detection, so queries had to scan more matches. In addition, since we did not prune stopwords, queries with multiple stopwords accumulated much larger postings lists, which increased runtime. Finally, longer queries expanded into many unigrams, bigrams, and trigrams, which increased the number of posting lists we touched and contributed to higher runtimes for those cases. The results of these bad queries also tended to be less relevant, since ranking was strictly TF-IDF/zone without any link-analysis information.

Optimizations:

By systematically testing the queries and deliberately focusing on those that initially performed poorly, we were able to improve ranking quality for ambiguous, phrase-like, and anchor-text-dependent queries. Through this refinement process, we reduced redundancy using near-duplicate detection while simultaneously maintaining or improving runtime for both easy and difficult queries through careful posting-list and top-k optimizations. We implemented duplicate detection, added PageRank and HITS link-analysis signals on top of our existing TF-IDF scoring, added 2-gram and 3-gram indexing, added positional data in the index, added anchor text counts into postings and weighted anchor rich documents higher, and created a web interface. Overall, the final search engine has been optimized to decrease run time and increase relevance.